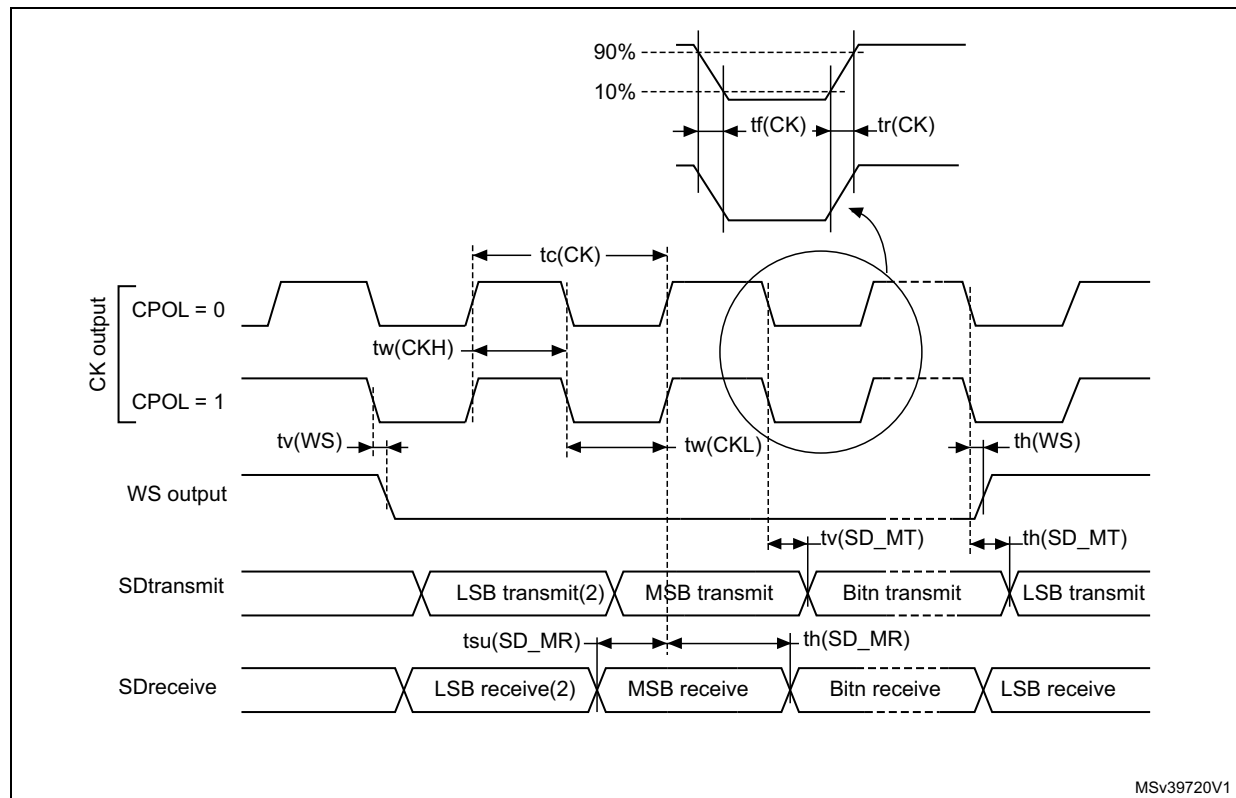


Figure 34. I²S master timing diagram (Philips protocol)

MSv39720V1

1. Evaluated by characterization. Not tested in production.
2. LSB transmit/receive of the previously transmitted byte. No LSB transmit/receive is sent before the first byte.

USB characteristics

The following table provides the electrical characteristics of the USB peripheral.

Table 67. USB electrical characteristics

Symbol	Parameter ⁽¹⁾⁽²⁾	Conditions	Min	Typ	Max	Unit
V_{DDIO2}	USB full speed transceiver operating voltage	-	3.0 ⁽³⁾	-	3.6	V
$t_{STARTUP}$	USB transceiver startup time	-	-	-	1	μ s
R_{PD}	Pull down resistor on PA11 (USB_DM), PA12 (USB_DP)	$V_{IN} = V_{DD}$	14.25	-	24.8	k Ω
R_{PU}	Pull Up Resistor on PA12 (USB_DP)	$V_{IN} = V_{SS}$, during idle	0.9	1.25	1.575	k Ω
		$V_{IN} = V_{SS}$, during reception	1.425	2.25	3.09	
VCRS	Output signal crossover voltage	-	1.3	-	2	V
ZDRV	Output driver impedance ⁽⁴⁾	Driving high or low	28	36	44	Ω